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The motion and frequency of large disturbance waves in annular two-phase flow of air-water mixtures

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Abstract—In annular two-phase flow large disturbance waves are nearly always observed. These travel rapidly along the surface of the liquid film and have a "milky" appearance as a result of light scattering by their extremely ruffled surface. The conditions under which these waves occur have been examined and measurements made of the wave velocity, separation and frequency.

The formation of the waves appears to coincide with a transition in entrainment observed in earlier work. Measurements of wave movements made by means of a ciné film and by a conductance probe method are in satisfactory agreement. The mean velocity of the disturbance waves was about 4–10 ft/sec compared with an air velocity of 40–150 ft/sec and a mean water velocity in the liquid film of 1–3 ft/sec. Mean wave velocity increased rapidly with increasing air flow but was insensitive to changes in water rate. The frequency of the waves was insensitive to air rate but increased in proportion to the liquid rate, thus suggesting that the waves are transporting the liquid. For any set of conditions the wave velocities measured are not evenly distributed about the mean and a number of peaks occur in the histogram. These peaks appear in the same positions for a number of different flow rates and the existence of "preferred velocities" is therefore possible. Similar, but more equivocal, observations are made with respect to wave separation.

INTRODUCTION

Two basic types of liquid film surface disturbance can be seen in annular two-phase flow, as has been noted in earlier work [1–3]. These will be designated "ripple" waves and "disturbance" waves. The first type correspond in appearance to the ripples commonly observed in falling films; they have a fairly small distance ($\frac{1}{4}$–2 in.) between peaks and travel at a relatively low velocity (0–5 ft/sec). The second type generally has a higher velocity (4–15 ft/sec), a much larger amplitude [1] and also a quite different appearance.

The most characteristic property of the disturbance waves is their circumferential coherence and their continuity of existence along the tube; "ripple" waves are overtaken by a disturbance wave and new ripples form in its wake, and hence ripple waves do not have a continuous life; nor do they usually form complete rings, though this can happen at low liquid and gas rates. Besides always forming rings, disturbance waves characteristically have a "milky" appearance; examination of high speed (18,000 frames/sec) ciné film [3] revealed that this appearance was due to the wave being covered by a pattern of wavelets, the highly ruffled surface causing light to be scattered. The ciné films also revealed that just before a disturbance wave arrived at a point the number of droplet impingements with the surface increased greatly, the droplets giving streaks or ripple rings. This latter fact suggested that the disturbance wave was responsible for initiating the entrainment of liquid into the gas core.

Fig. 1 is an attempt to illustrate the difference between the two types of wave.

In the work to be described below the object was to study the disturbance waves. The main part of this work, using a photographic method, was carried out during the summer of 1961 when one of us† was a vacation student at Harwell. Owing to limitations of available time the range of flow rate studied is rather small and the flow rates themselves are somewhat low. Despite this, the data provide some very useful preliminary indications of the phenomena involved. Subsequently,

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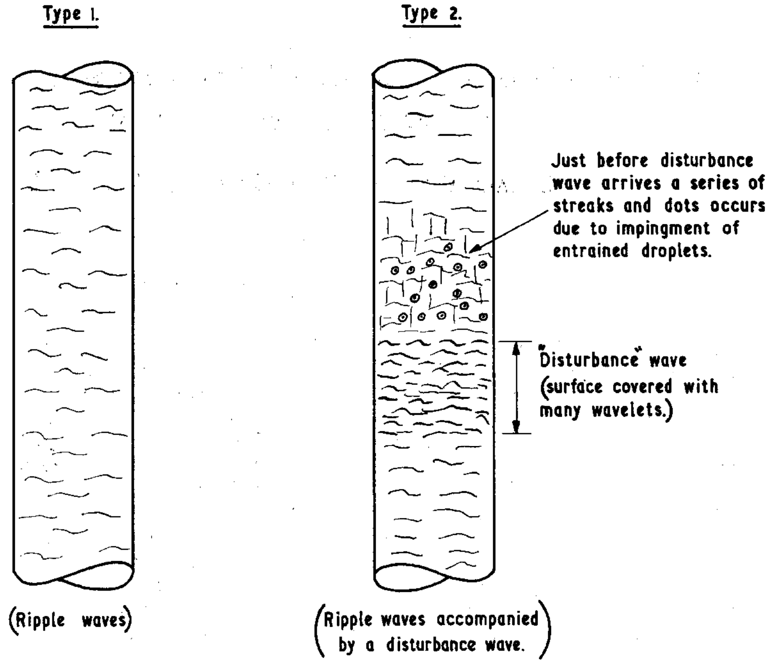


FIG. 1. Basic types of disturbance on liquid film surface in annular flow.

for the sake of completeness, pressure drop and film thickness were measured, and also wave frequency and velocity, using conductance probe techniques [1], for the same rates of flow.

EXPERIMENTAL PROCEDURE

The apparatus used was similar to that devised for earlier experiments on air/water systems, which have been described in detail elsewhere [1, 4]. A long, vertical Perspex tube, $1\frac{1}{4}$ in. bore, was fed with metered supplies of air and water; the air from the site mains, via an orifice plate, and the water from a sump tank, recirculated by a centrifugal pump, via rotameters.

A sketch of the flow tube is given in Fig. 2. The air entered at the bottom of the tube and the water was injected around the whole periphery via a porous sinter injector. At the exit the two-phase mixture was passed into a cyclone separator from which the air was discharged to atmosphere and the water returned to the sump tank. A small amount of sodium chloride was added to the water to allow the film thickness to be determined [1]. The positions of pressure tappings, film thickness probes and flanged joints along the flow tube are shown, the total length from the injector to the final flange being 22 ft.

The porous sinter injector, Fig. 3, had been developed previously in connection with work on entrainment [5]. It has the advantage that it gives little or no adventitious entrainment at the inlet, and it was chosen for the present study for this reason and to minimize the effect of flow tube pressure oscillation on the inlet water rate. This latter property avoids spurious waves which might be derived from oscillations imposed on the injector.

Pressure drops were measured between tappings 1 and 2, 2 and 3, and 3 and 4, using water-purged lines, and film thicknesses were measured, at five locations, by the method described in Ref. [1]; (the data given here, however, may have been somewhat less reliable owing to capacitance effects in long leads).

Determination of flow regimes

As a preliminary to the wave motion studies a survey was made of the various flow regimes in the tube, the following being identifiable:

(a) Slug and churn flow: the air flows in slugs, or the slugs break down and the liquid undergoes a vertical oscillatory motion.

(b) Annular flow: this region was divided into four categories:

(i) Non-wetting region: the water fails to wet the whole tube surface. Patches of rippling liquid film occur over certain sections of the tube: in between, the water flows over the dry surface in streams or drops.

(ii) Region of ripple waves only: the surface is covered by a complete film with ordinary waves but no disturbance waves.

(iii) Disturbance wave region: here there are disturbance waves superimposed on the ripple waves. An attempt was made to distinguish between cases where the ripple waves are large and where they are much smaller. The disturbance waves do not exist immediately above the injector but grow

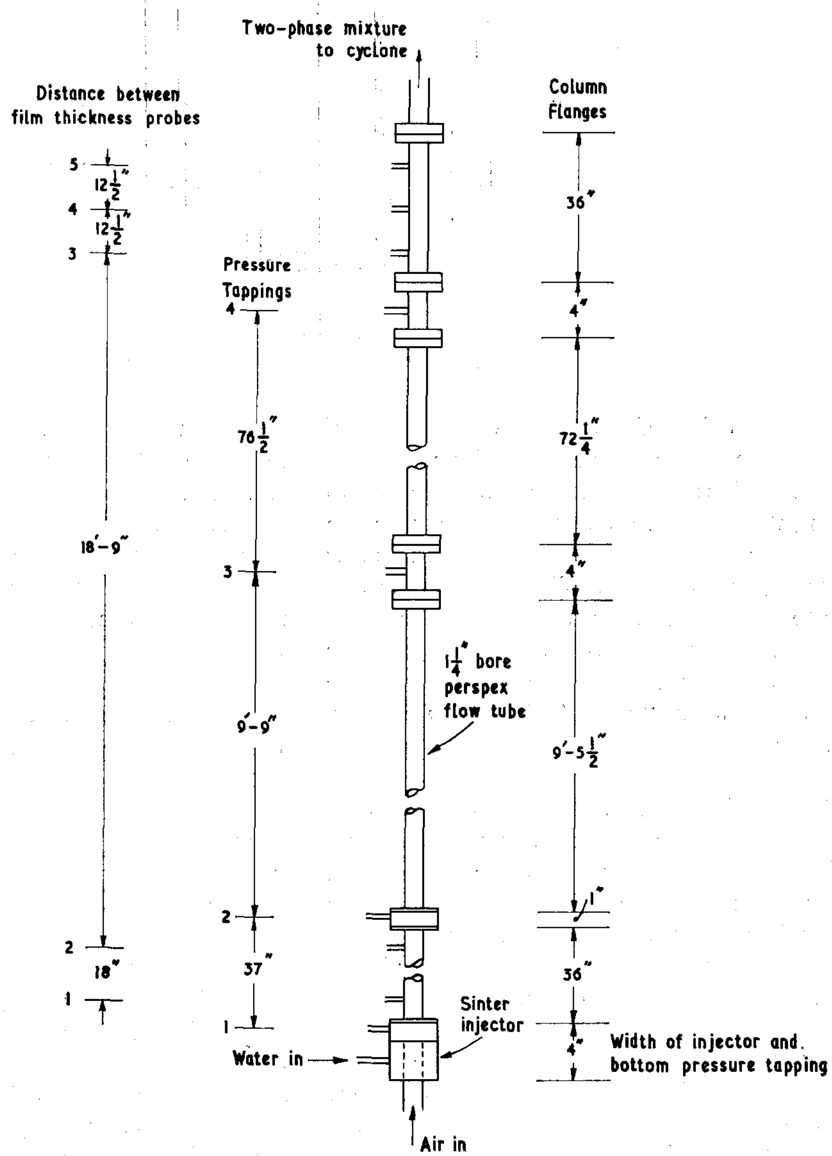


FIG. 2. Flow tube layout.

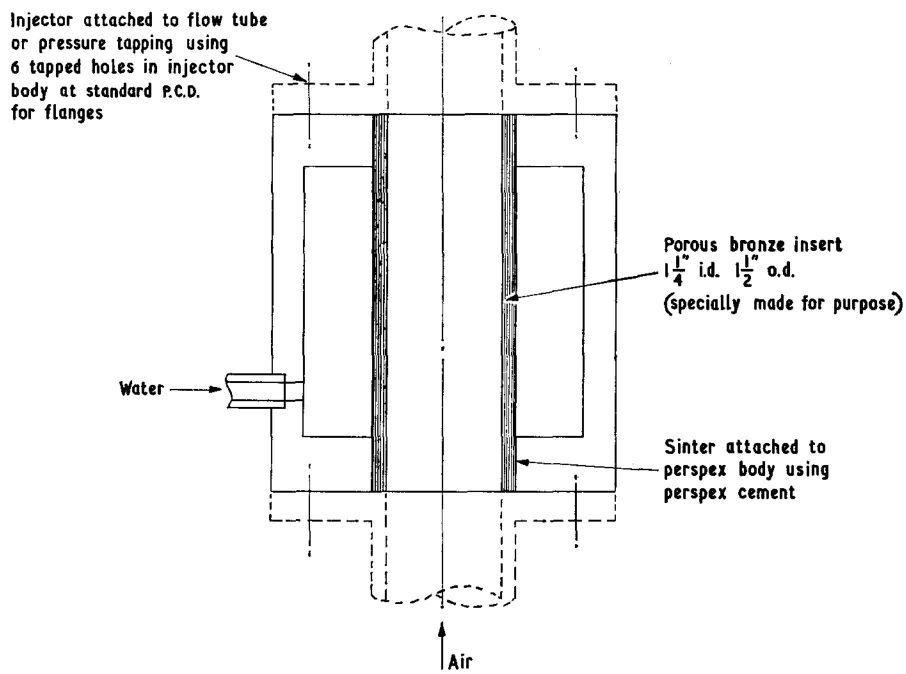


FIG. 3. Porous bronze sinter injector.

in the first few feet of the tube. After this initial region they appear to have a stable form.

(iv) Pulse region: this is an intermediate region between (ii) and (iii) above. Disturbance waves do occur but are not formed continuously and appear to come in groups or "pulses". This is unlikely to be due to oscillation in the water input rate but is more probably analogous to the intermediate region between laminar and turbulent single phase flow in a tube, where "pulses" of turbulence occur, and may be related to resonances in the gas inlet system.

(c) Intermediate region: In between the slug/churn region and the annular flow region a somewhat indeterminate area exists where the liquid is mainly adjacent to the walls but oscillations in the liquid surface occur which are rather more violent than those found in annular flow though not oscillating bodily as in churn flow.

Ciné film method

The light scattering property of the disturbance waves was utilized in studying their motion. The flow tube was illuminated tangentially and photographed at 32 frames/sec by means of a 35 mm ciné camera mounted some 15 ft away from the tube and about 10 ft from the ground. By turning the camera through an angle it was possible to arrange the image across a diagonal of each frame and thus include nearly the whole length. A timer was constructed, consisting of an arm rotated by means of a synchronous motor against a marked card. This was mounted adjacent to the tube and served as a check on the framing rate of the camera: in fact, after an initial camera acceleration period which was discarded before the film was analysed, the camera speed was found to be constant and almost precisely 32 frame/sec. A black board was mounted adjacent to the tube on which white distance markers were taped at 1 ft intervals.

After development the negative films were projected from a single-framing projector via a mirror onto a horizontal white card on a table. The positions of the large waves, which could be clearly discerned after some practice, were noted and their positions in successive frames recorded. Two operators were employed—one noted the position of the wave on the projected image using a specially calibrated rule and the other plotted the position on sheets of graph paper. Even with the limited range of flow rates covered some 10,000 such points had to be plotted.

Conductance probe method

Conductance probes allowed film thickness traces to be obtained, from which the time of transit of a wave could be computed. The pairs of probe electrodes were $\frac{1}{8}$ in. diameter, $\frac{1}{2}$ in. apart, mounted flush with the tube surface, and the conductivity of the film between them was used to modulate a 10 kc/s carrier current which was displayed on an oscilloscope, the envelope indicating the film thickness. By a method described elsewhere [1] up to four half-envelopes could be displayed simultaneously: three are shown in Fig. 9, derived from the three probes of Fig. 2. The use of large probe geometry generally ensured that the large waves could be effectively differentiated from the ripples.

Results

The regimes of flow

The results for flow regimes are plotted in Fig. 4. The various types of flow occur in well defined areas on the "flow map". The first interesting point which arises is that the transitions from slug and churn flow to "intermediate flow", and from

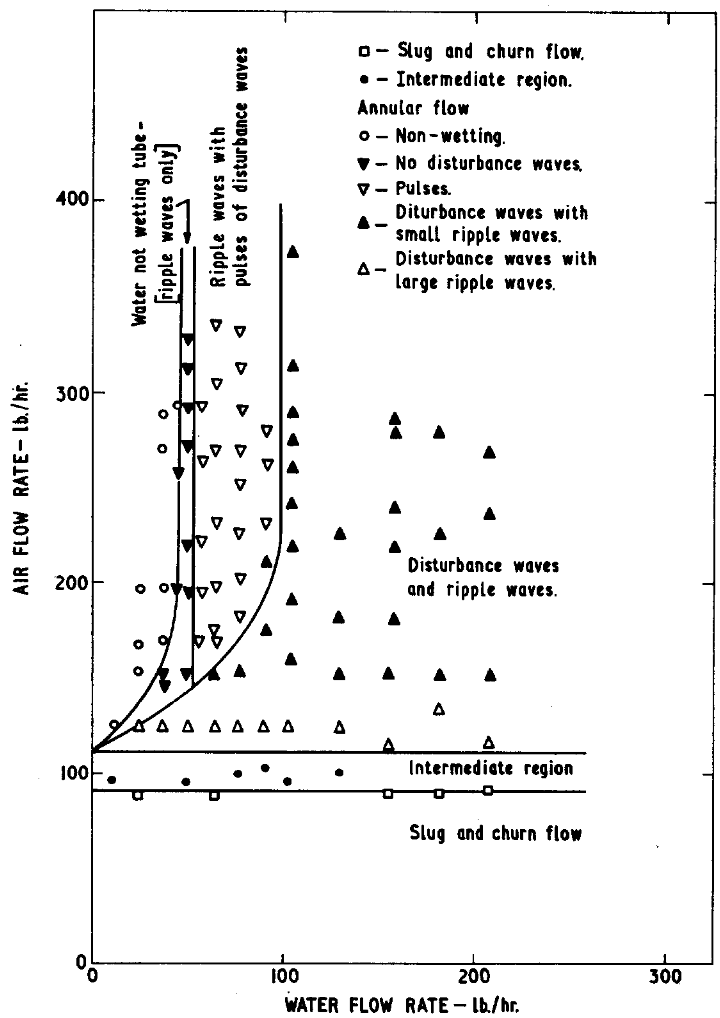


FIG. 4. Regimes of flow in $1\frac{1}{4}$ in. bore tube.

"intermediate flow" to annular flow, occur at constant air rates and are independent, within the range studied, of the water rate. This confirms the observation of GOVIER *et al.* [6] and the theoretical predictions of WALLIS [7] and NICKLIN [8].

In the annular regime the type of flow passes through several transitions as the liquid rate is increased. Considering first of all the higher air rate region, the changes from non-wetting to wetting, from "ripple waves only" to the pulse region, and from pulses to disturbance waves, all appear to take place at constant liquid rates.

There is strong evidence that the transition to disturbance waves is accompanied by the initiation of entrainment. In the work reported in Ref. [9] it was found that above a certain value of y_i^+ (the dimensionless distance parameter at the interface), the entrainment was predicted by the correlation of WICKS and DUKLER [10]; below this value of y_i^+ the entrainment is less than predicted. The critical value of y_i^+ was in the range 16–18‡. If one assumes that the entrainment is zero—a reasonable assumption for water rates below about 100 lb/hr introduced through a porous sinter injector—then it follows that at a constant liquid rate a constant value of y_i^+ will be observed. Making this assumption and referring to the Table in Ref. [11] relating the dimensionless film flow rate with y_i^+, it follows that the transition from "ripple waves only" to the pulse regime occurs at a y_i^+ of about 12 and that the transition from pulses to the fully developed disturbance wave region occurs at $y_i^+ \sim 18$. The close relationship between these figures and those observed for the transition in the entertainment strongly suggests that the disturbance waves are the source of the entrainment. At lower air rates the water rate for the transitions falls and eventually reaches zero at the boundary with the "intermediate" regime. The phenomenon of annular "flow" at zero net water flow rate would seem, at first sight, to be impossible but it has been clearly observed in the present tests. A stationary film is supported on the wall; the water presumably flows down the film under gravity and is dragged up again to the top by the air in waves of the "disturbance" type. This condition can be very stable. The possibility of liquid transport in the waves in this condition of stationary film may support the idea of liquid transport by waves in ordinary annular flow which is strongly suggested by the results on wave motion to be presented below.

The transition from the non-wetting to the wetting condition has been found to depend highly on the exact condition of the surface of the tube. The Perspex tubes seem to wet better after being used for some time. The tube dries out from the injector or from the outlet.

Pressure drop and film thickness measurements

A series of air and water rates was chosen for study in the disturbance wave region and at these flow rates the pressure drop and film thickness were measured and the results are tabulated in Table 1. The film thickness for any given set of flow rates was found not to vary much along the tube (though the set of probes immediately above the injector did tend to give rather higher values). In Table 1 the average value of film thickness is given. Pressure gradients appeared to fall continuously along the tube but the effect is relatively small (10 per cent in 20 ft). No attempt has been made to analyse the film thickness and pressure drop data; they were obtained for reference purposes only and the flow rate ranges have been adequately covered in earlier and concurrent work aimed specifically at pressure drop and liquid film thickness studies.

Wave motion results—photographic method

The general pattern

The results from the measurements on the ciné films were plotted directly on to graphs of distance versus time, the first 8 ft. of the tube being ignored. The distance/time plot for each disturbance wave was found, in general, to be a straight line, indicating a constant velocity for the waves. The slope of the line gives the velocity and the wave separation can be obtained by estimating the distance between the line and its nearest neighbours. In a few cases anomalous behaviour was observed, which is discussed below. Typical sections of the plots

‡ y^+_i is defined by $y^+_i = u^* \rho_L y_i / \mu_L$ where ρ_L and μ_L are the liquid density and viscosity respectively and y_i the film thickness. The friction velocity u^* is obtained from $u^* = \sqrt{(\tau_0/\rho_L)}$ where τ_0 is the shear stress at the wall.

Table 1. Pressure drop and film thickness data

Run Number	Flow Rates: Air (lb/hr)	Flow Rates: Water (lb/hr)	Average Film Thickness in. × 10^{-3}	Pressure Drop 1–2 (lb/ft² ft)	Pressure Drop 2–3 (lb/ft² ft)	Pressure Drop 3–4 (lb/ft² ft)	Inlet Pressure (p.s.i.a.)
12	148	65	10·52	3·10	2·99	2·67	16·44
13	147	90	11·60	3·60	3·43	3·23	16·63
14	148	116	12·16	4·02	3·95	3·74	17·08
15	147	143	12·84	4·49	4·44	3·95	17·20
16	144	169	13·85	5·08	4·73	4·54	17·39
17	147	209	14·81	5·56	5·08	5·02	17·71
18	124	103	13·72	3·95	3·76	3·83	16·39
19	203	103	9·55	4·92	4·41	4·07	18·08
21	278	103	7·43	6·27	5·81	5·02	20·31
22	304	103	6·95	6·88	6·51	5·67	20·90
23	373	103	5·91	8·47	7·80	7·23	22·99
24	147	209	14·32	5·86	5·22	4·85	17·72
25	198	209	11·41	6·36	5·91	5·42	19·30
26	223	209	10·44	6·70	6·55	6·10	20·17
27	218	90	8·17	4·31	4·13	4·17	18·21
28	218	130	9·28	5·12	5·18	4·86	18·91
29	218	156	9·46	5·81	5·50	5·25	19·23
30	218	183	10·06	6·41	5·92	5·47	19·54

obtained are given in Figs. 5–8. It can be seen from these figures that:

(a) At constant flow rates, although each wave has a constant velocity, not all waves have the same velocity.

(b) As the liquid rate is increased at constant air rate the spacing between the disturbance waves is considerably reduced (c.f. Figs. 5 and 6) but the velocity remains approximately constant,

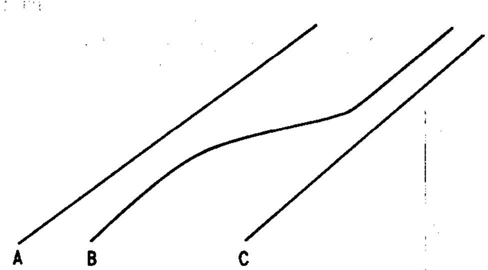
(c) As the air rate is increased the wave velocity increases (c.f. Figs. 7 and 8).

These observations will be more clearly shown in the quantitative analysis of the results given alter.

Exceptions to the general pattern

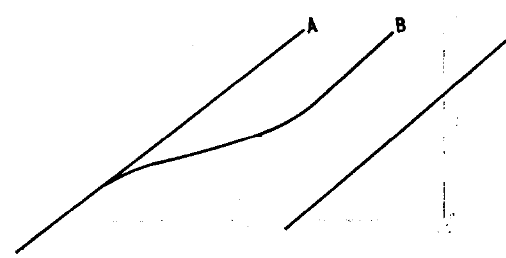
A number of types of anomalous behaviour of the disturbance waves were observed. These exceptions to the general pattern occurred most frequently at lower air rates but are by no means confined to this region. The anomalies are listed and described below in the order of their most frequent occurence, (the frequencies being listed in Table 2).

(a) *Shift of wave position.* In some cases the wave underwent a fairly sharp jump from one line of motion to another:



Wave B moves, by means of a deceleration and acceleration mechanism, from a position close to the wave above it (A) to a position close to the wave below it (C). It is of interest to note that no case was observed of a wave moving from a position close to C to one close to A.

(b) *Wave Splitting.*



Wave A splits and part of it proceeds without any deviation from its original course. The other part

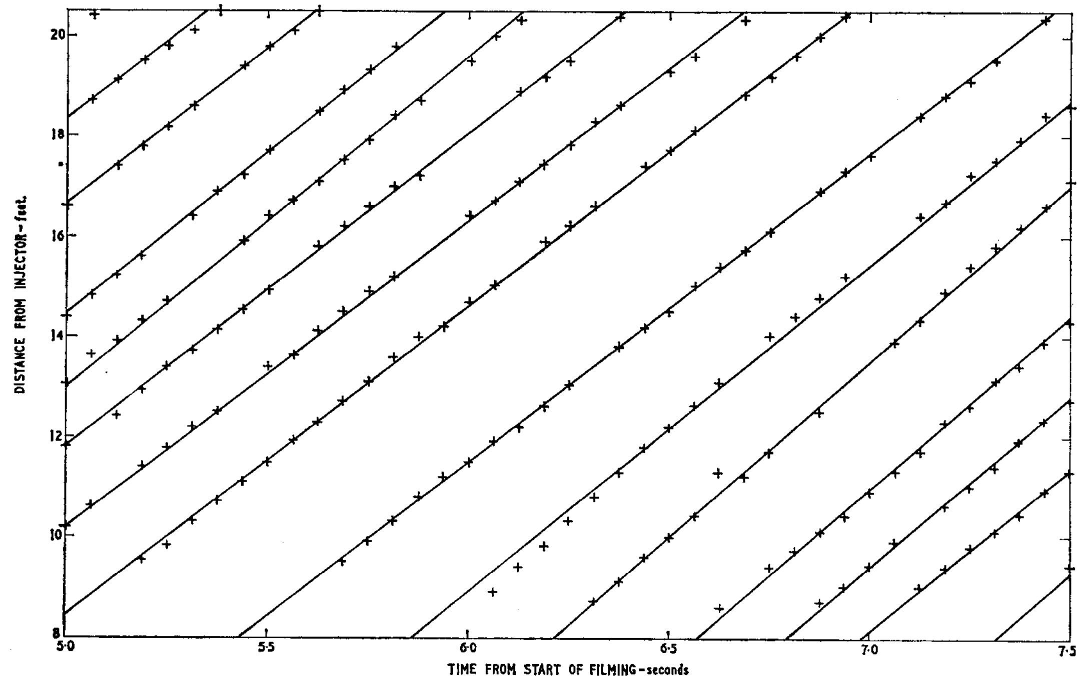


FIG. 5. Time/distance plot from ciné method, Run 19, water rate 103 lb/hr, air rate 203 lb/hr.

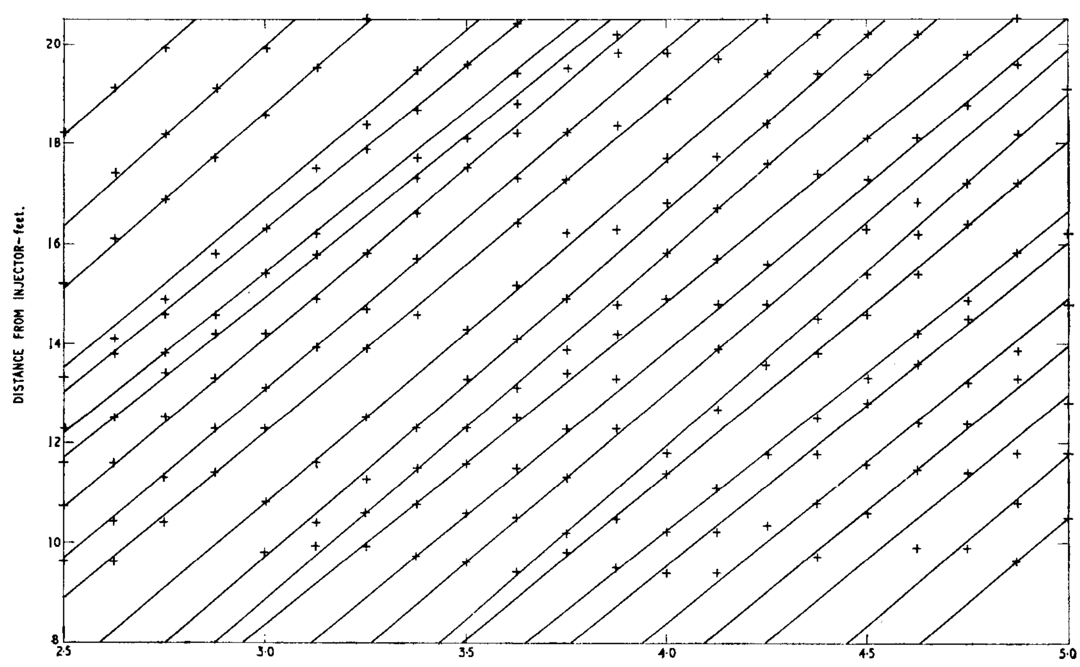


FIG. 6. Time/distance plot from ciné method, Run 25, water rate 209 lb/hr, air rate 198 lb/hr.

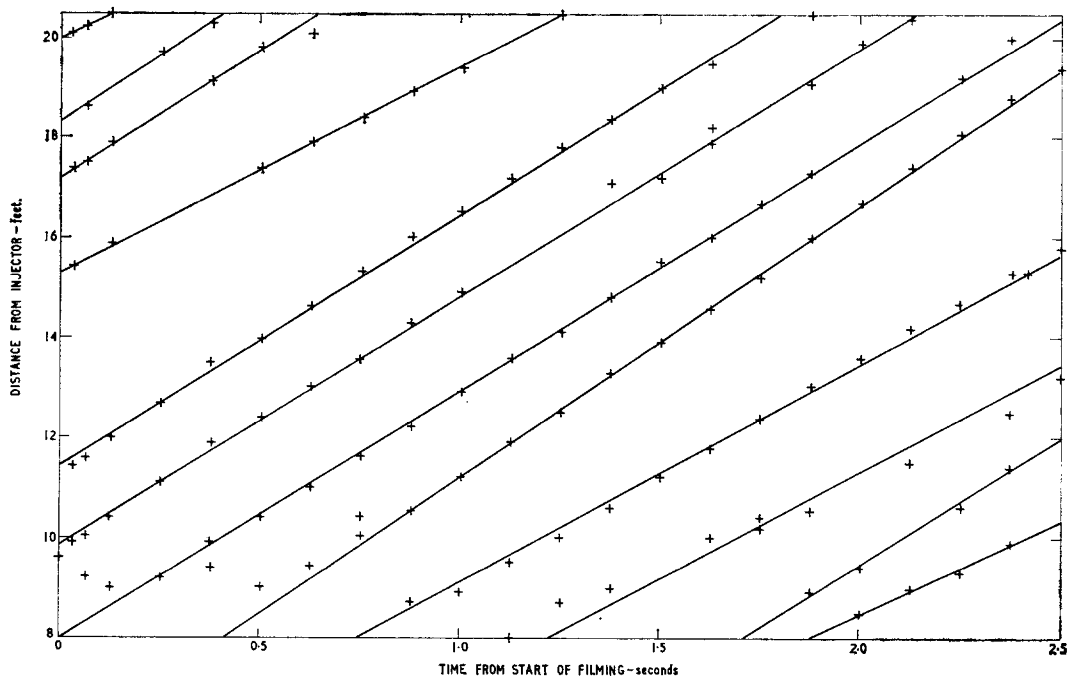


FIG. 7. Time/distance plot from ciné method, Run 13, water rate 88 lb/hr, air rate 147 lb/hr.

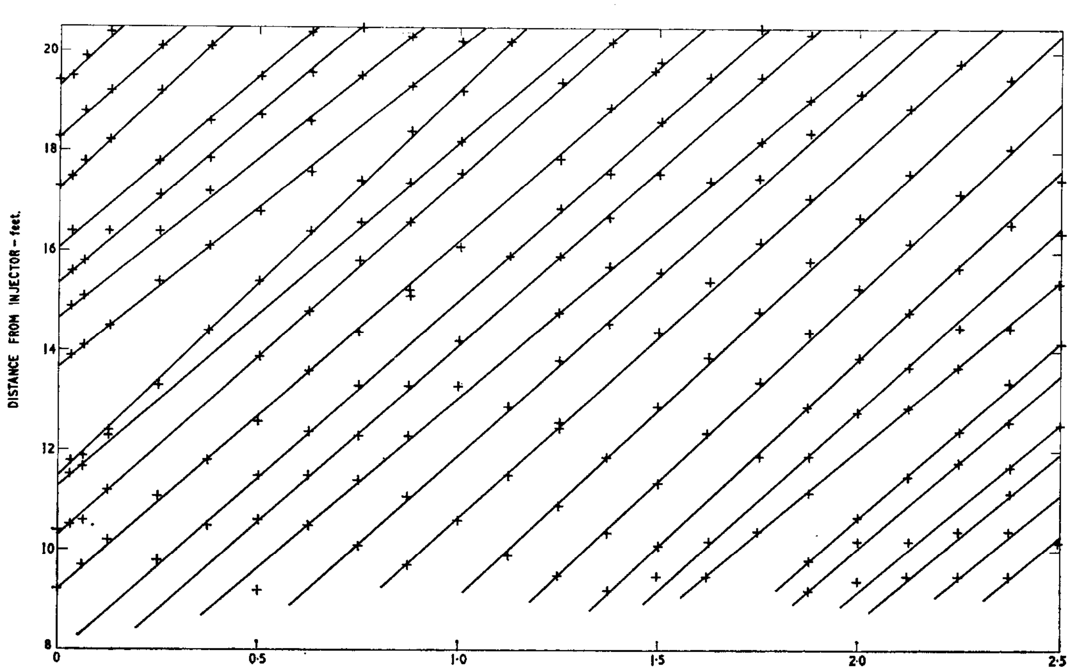


FIG. 8. Time/distance plot from ciné method, Run 30, water rate 183 lb/hr, air rate 323 lb/hr.

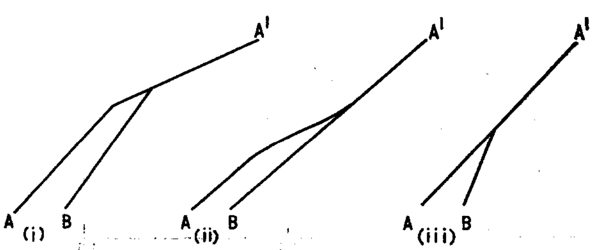
Table 2. Wave motion results

Run Number	*Air flow rate* (lb/hr)	*Water flow rate* (lb/hr)	*Photographic method*											*Conductance probe method*		
							No. of occurences of anomalous behaviour									
			No. of waves meas.	*Mean wave velo-city* (ft/sec)	*Mean wave sepn. dis-tance* (ft)	*Mean sepn. time* (sec)	*Shift of posi-tion*	*Wave split-ting*	*Wave merging Type* (i)	*Type* (ii)	*Type* (iii)	*Shift of position of two waves Type* (i)	*Type* (iii)	*No. of waves meas.*	*Mean wave velo-city* (ft/sec)	*Mean wave sepn time* (sec)
12	148	65	16	5·18	3·87	0·750	—	2	—	—	—	—	—	12	4·63	0 427
13	147	90	31	4·91	2·02	0 421	2	—	1	—	—	—	—	22	4·72	*
14	148	116	43	4·89	1·52	0·317	1	—	—	—	—	1	—	21	4·83	*
15	147	143	47	4·94	1·35	0·275	5	—	—	1	1	—	—	—	*	*
16	144	169	26	5·31	1·35	0·259	6	—	—	1	—	—	1	—	*	*
17	147	209	33	5·60	1·22	0·225	—	—	—	1	—	—	—	—	*	*
18	124	103	30	4·29	1·48	0·354	1	1	—	1	—	—	—	—	*	*
19	203	103	35	6·43	1·81	0·293	1	—	—	—	—	—	—	38	6·23	0·254
21	278	103	42	7·88	1·89	0·239	—	—	—	—	—	—	—	59	7·71	0·176
22	304	103	29	8·33	1·69	0·206	—	—	—	—	—	—	—	60	8·22	0·166
23	373	103	33	9·49	1·54	0·179	—	—	—	—	—	—	—	51	8·96	0·175
25	198	209	50	6·85	1·12	0·172	—	—	—	—	—	—	—	58	7·14	0·167
26	223	209	57	7·27	1·02	0·142	—	—	—	—	—	—	—	71	7·38	0·136
27	218	90	26	6·85	2·16	0·334	—	—	—	—	—	—	—	25	6·66	0·278
28	218	130	51	6·51	1·17	0·180	2	1	—	—	—	—	—	29	6·78	0·170
29	218	156	45	7·02	1·28	0·182	—	—	—	—	—	—	—	—	—	—
30	218	183	51	6·90	1·03	0·151	1	—	—	—	1	—	—	23	7·19	0·150
31	367	130	49	9·26	1·32	0·143	—	—	—	—	—	—	—	—	—	—

* No measurement possible.

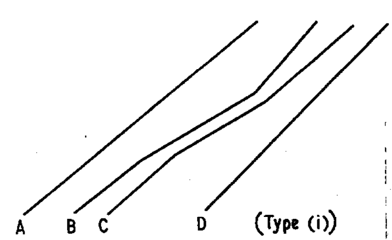
(B), however, decelerates and then accelerates so that it is again moving at approximately the same velocity as A, but now at a position approximately midway between A and C.

(c) *Wave merging*. Since there are different disturbance wave velocities in the system it is not surprising that the waves will occasionally collide. Three types of collision were observed:



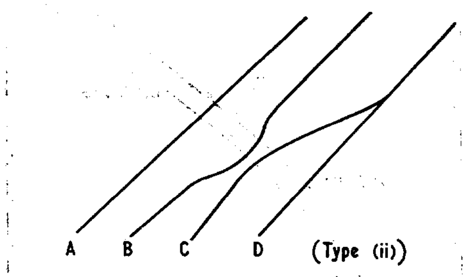
It is interesting to note that in no case does the collision result in the combined wave having a higher velocity than its component. Subsequent high speed ciné films have revealed that such collisions are accompanied by a burst of extra entrainment.

(d) *Shift of position of two waves*. This phenomenon is a complex example of type (a):



In the first case (type i) wave B shifts in the manner described above but wave C also undergoes a shift and the net result is a movement of waves B and C towards wave D. Only one case of this type was observed.

Another form of this behaviour is as follows:



In this case wave B performs a deceleration and moves towards wave C. In the meantime wave C also shifts and combines with wave D. Wave B then shifts back to a position close to its original path-line. Again, only one case of this type of behaviour was observed.

Computer analysis of the results

The treatment of the data obtained, even for the limited range of variables studied, presented an almost impossible task for hand processing and a computer was therefore used. The objects in analysing the results were as follows:

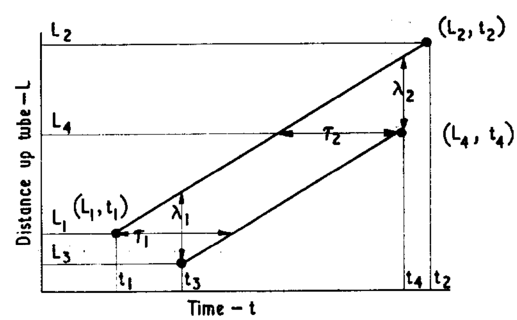
(i) To obtain the velocity of each wave and its separation in time and distance from the preceding and succeeding waves.

(ii) To obtain the mean time and distance separation and the mean velocity of the set of waves.

(iii) To derive histograms of the distribution of velocities in the set of waves and of the distribution of time and distance separation.

The information from the original time/distance plots was condensed by feeding into the computer the time/distance co-ordinates of the two points where the wave was first and last observed. From these co-ordinates the mean velocity is readily calculated and, by storing the co-ordinate for each line in the machine the initial and final time and distance separation from the preceding and succeeding wave was calculated. The exceptional waves—described above—were treated arbitrarily by taking a mean straight line path in the case of wave shifts and by treating merged waves and split waves as individual waves up to and from the point of merging or splitting. Since the anomalous waves represented only 4 per cent of the total number the arbitrary method of treating them did not have any large effect on the final result.

In order to obtain a histogram of velocity the computer scanned the measured velocities for the set of waves and found the minimum and the maximum velocity. The interval between minimum and maximum was then divided into ten parts and the number of waves in each increment counted and the histogram elements calculated. By selecting the intervals in this way, rather than fixing them, the "preselection" of the peaks in the histogram curves is avoided. A similar procedure was used for the time separation and distance separation, the interval between maximum and minimum being divided into twenty parts. The time and distance separation between any pair of successive waves is constantly varying and this raised some difficulty. Consider the time-distance plots of two waves as shown in the sketch below. It will be noted that the waves are being measured in the tube *simultaneously* only between times t_3 and t_4 and it is therefore only between times t_3 and t_4 that the distance between the waves may be properly measured.



Similarly the time between waves may only properly be evaluated in the length range L_1 to L_4. It is obvious that τ, the time between waves, varies linearly with length in the range $\tau = \tau_1$ at $L = L_1$ and $\tau = \tau_2$ at $L = L_4$ and similarly the wave separation distance varies linearly between λ_1 at $t = t_3$ and λ_2 at $t = t_4$.

The twenty histogram cells between the maximum and minimum wave time separation were accumulated in terms of the total distance over which the time separation range in the cell was observed. The histogram cells for distance separation were accumulated in terms of total *time* spent in each distance range.

Effect of phase flow rate on wave velocity and wave separation.

The results for mean velocity, mean time separation and mean distance separation are given in Table 2. In principle, of course, the product of the mean velocity and the mean separation time should equal the mean separation distance. In fact, however, the method used for obtaining the means of separation time and distance can lead to directly computed values of mean separation distance rather different from the product of mean velocity and mean separation time. For the results presented here the latter are about 1–3 per cent higher than the former. This might suggest a slight overall tendency of the waves to converge as they pass up the tube.

Fig. 10 shows the effect of water rate on average disturbance wave velocity at two constant air rates. The velocity is substantially independent of water rate in the range studied though there is some variation with a suggestion of a flat minimum. In Fig. 11 mean wave velocity is plotted against air rate. Wave velocity increases sharply with increasing air rate. A power law represents the data well, wave velocity being proportional to air rate to the 0·75 power.

In Figs. 12 and 13 mean separation time (i.e. the reciprocal of mean frequency) is plotted against air rate and water rate for constant water and air rate respectively. The mean time separation decreases with both air and liquid rate; in the case of increase of air rate at constant water rate the decrease is linear and in the case of increase of water rate at constant air rate a hyperbolic decrease in time separation is suggested. The decrease in time separation with increasing air rate at constant water rate is mainly due to the increase in wave velocity and the distance between the waves is not affected greatly (see Table 2). Since wave velocity does not depend greatly on water rate, however, the decrease in separation distance with increasing water rate is similar to that for separation time. The hyperbolic nature of the curves in Fig. 12 suggested a constant product of water rate and separation time; this was tested and the results are presented in Table 3 below.

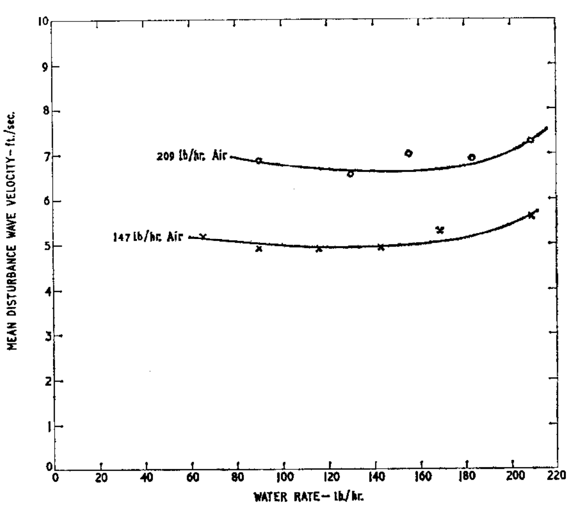


FIG. 10. Effect of water rate on mean disturbance wave velocity at constant air flow rate.

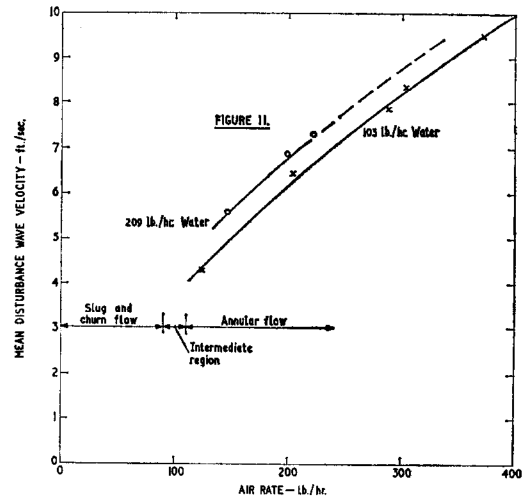


FIG. 11. Effect of air rate on mean disturbance wave velocity at constant water flow rate.

Table 3

Air rate 147 lb/hr						
Water rate, W (lb/hr)	65	90	116	143	169	209
Separation time, τ (sec)	0·750	0·421	0·317	0·275	0·259	0·225
$W\tau$	48·8	37·9	36·8	39·3	43·8	47·1
			Average value of $W\tau = 42{\cdot}3$			
Air rate 218 lb/hr						
Water rate, W (lb/hr)	90	130	156	183	209	
Separation time, τ (sec)	0·336	0·180	0·182	0·151	0·142	
$W\tau$	29·1	23·4	28·4	27·6	29·7	
			Average value of $W\tau = 29{\cdot}7$			

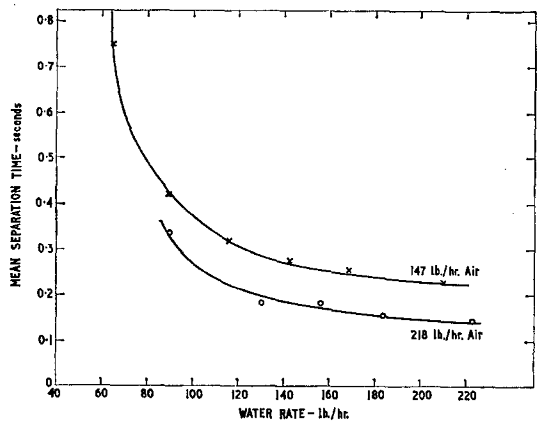


FIG. 12. Effect of water rate on mean time separation and disturbance waves at constant air rate.

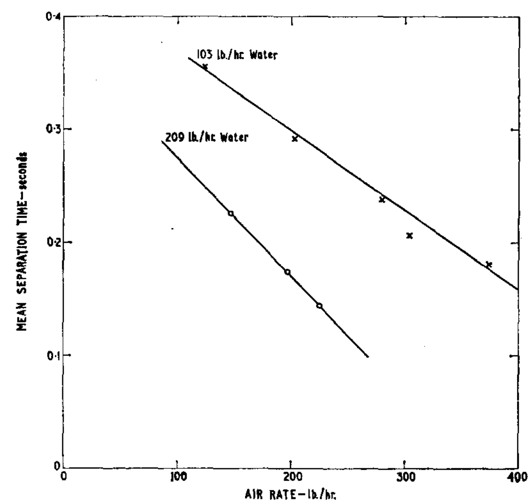


FIG. 13. Effect of air rate on mean time separation of disturbance waves at constant water rate.

As will be seen the value of $W\tau$ is fairly constant over the range of values of liquid rate. This indicates that as the liquid rate is increased, at constant air rate, although the velocity of the waves remains roughly constant their frequency increases in rough proportion to the liquid rate.§ The reason for this may be that the waves are transporting fluid and, whereas they remain of roughly the same size with increasing water rate, their number increases to accommodate the extra flow. It is interesting to make an approximate calculation of the amplitude of the waves assuming that all the flow occurs in them. The total length of the disturbance area associated with each wave is ½–3 in. Assuming a length of 1 in. and an isosceles triangle shaped wave then the peak height for total liquid transport in the waves is readily calculated from $W\tau$ as about 0·16 inches for an air rate of 147 lb/hr and about 0·11 in. for an air rate of 218 lb/hr. These peak heights are rather higher (by a factor of 2–3) than would be suggested by conductance probe measurements [4]. The transport of liquid, therefore, may not be exclusively in the waves though their contribution may be of importance.

§ Recent work by NEDDERMAN and SHEARER [12] shows that this tendency does not continue when the (total) liquid flow rate is increased beyond the range used here.

Distribution of wave velocity and wave separation

The histograms of wave velocity and separation were derived as described earlier. They are, of course, discontinuous: the histogram functions $h(\tau)$, $h(v)$ and $h(\lambda)$ are defined in terms of the selected number of discreet intervals defined between the maximum and minimum values. For instance, the histogram for the distribution of velocity is defined

$$[h(v)]_{\frac{v_1+v_2}{2}} = \frac{\text{number of waves having velocities in range } v_1 \rightarrow v_2}{\text{total number of waves} \times (v_2 - v_1)} \tag{1}$$

$h(\tau)$ and $h(\lambda)$, the histogram functions for separation time and distance, were defined in a similar manner.

A typical histogram for velocity distribution is given in Fig. 14. It will be seen that an even distribution about a mean is not obtained. A number of peaks occur in the distribution and it was observed that the approximate positions of these peaks were recurrent from one test to another. This point is illustrated by the following table (see Table 4) which lists the position of the peaks for the series of tests at an air flow rate of 147 lb/hr.

It will be seen that the peaks occur at quite constant values from one test to another. The difference between mean peak values also seems to be constant at about 0·4 ft/sec. There is some

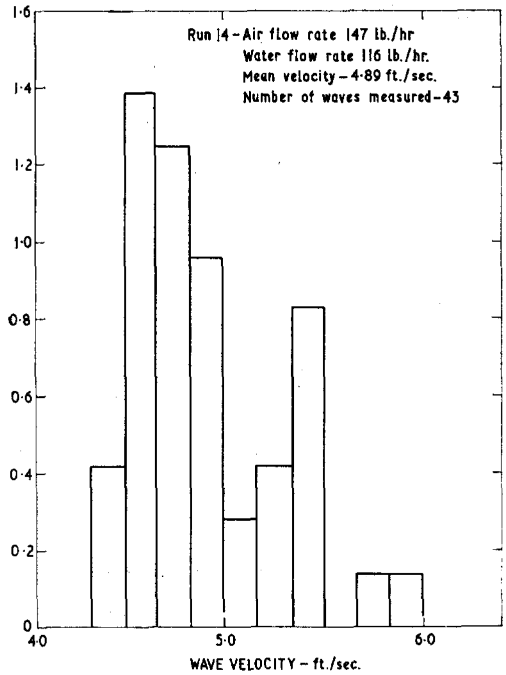


FIG. 14. Histogram of wave velocity.

suggestion that the peaks occur at similar values over the whole range of air flow rates though, as the air rate is increased, new and higher values of the velocity at the histogram peak are observed.

Table 4

Water rate (lb/hr)	Peak positions in velocity (ft/sec)					
65	4·3		4·9	5·3	5·8	6·2
90	4·2	4·6	4·9	5·35		
116		4·5		5·4	5·8	
143			4·9	5·4		
167		4·5		5·25	5·65	
209				5·25	5·8	6·2
Mean value of peak	4·25	4·5	4·9	5·3	5·75	6·2

Typical histograms of wave time and distance separation are given in Figs. 15 and 16. These histograms can be interpreted as being incomplete normal distributions (they have a high peak close to the mean). In this case again, however, secondary peaks occur and may indicate "preferred" values of separation though such a conclusion must be regarded as very tentative.

Wave motion results—conductance probe method
Limitations in method

The conductance probe method appears to fail rather badly when large ripples are present in addition to the disturbance waves. This failure is due to the fact that it is not possible, on the conductance probe record, to distinguish high amplitude ripples from the disturbance waves. In some cases a number of the disturbance waves can be recognised although it is impossible to say if all the waves have been measured—in these cases a measurement of wave velocity is possible but no estimate of wave frequency could be made. The large amplitude ripple region occurs at low air flows and high liquid flows. (In Table 2 the conditions under which measurements using the conductance probe method were not possible are indicated.) Traces which illustrate some of the above points are given in Fig. 9.

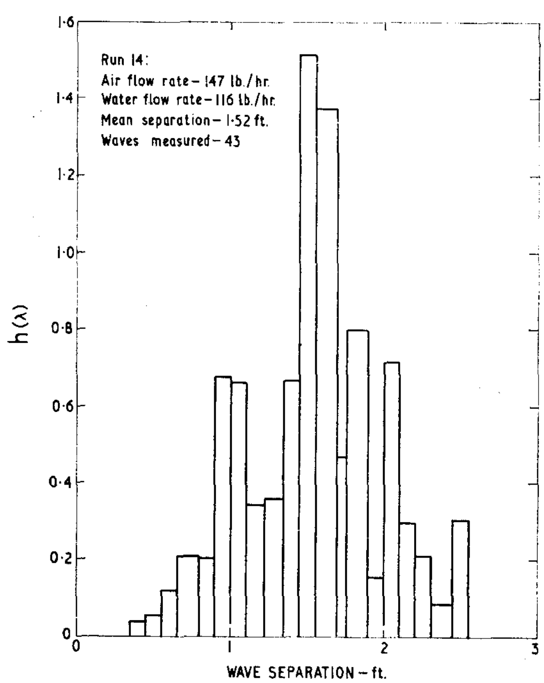


FIG. 15. Histogram of wave separation.

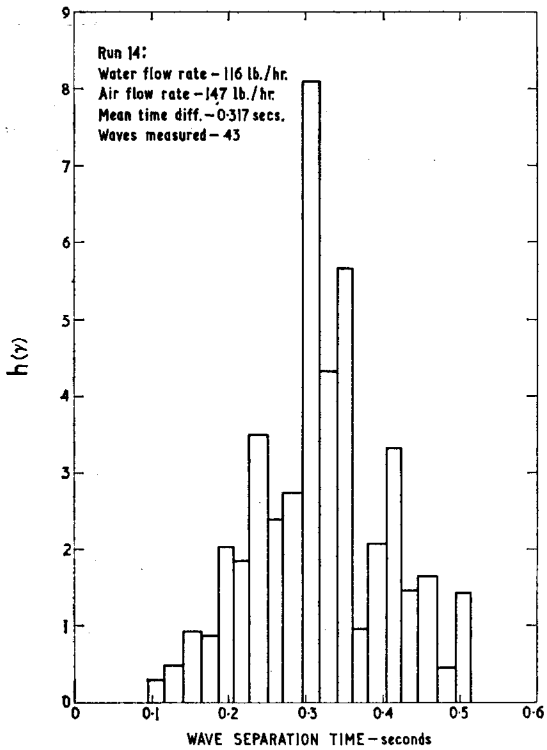


Fig. 16. Histogram of wave separation time.

Wave velocity and time separation

Wave velocity measurements were possible in the majority of the conditions studied and it will be seen, on reference to Table 2, that the mean velocity values obtained are in excellent agreement with those obtained by the ciné film method. The mean time separation values, on the other hand, are rather lower than the ones obtained by the ciné film. This small difference could be due to the fact that "double waves" (pairs of disturbance waves travelling in close proximity) are seen distinctly as two separate peaks when using the conductance probe method whereas they may not be discriminated on the ciné film. From this point of view, therefore, the conductance probe method is the more accurate but, in view of the limitations described above, the results from the ciné film measurements have been used in preparing all the graphs presented.

The extent of the agreement between the two methods of measurement, however, does show that high amplitudes are associated with the disturbance waves.

Discussion

Waves which are probably of similar type to the disturbance waves reported here have been observed in horizontal annular two-phase flow by Kinney *et al.* [13] and by Knuth [14]. Knuth [14] found that the initiation of "roll" waves (probably corresponding to the "disturbance" waves reported here) occurred at a constant liquid rate independent of gas rate; he also observed that the liquid rate for initiation was a function of liquid viscosity. Knuth correlated his results by plotting y_{iw}^+, the value of the friction distance parameter at the liquid/gas interface at which roll wave initiation starts, against the ratio of gas to liquid viscosity. It is doubtful if this relationship would be valid in application to vertical flow and, in any case, the results on which it is based are somewhat limited.

In more recent work Hanratty and Hershman [15] have discussed the transition to "roll" waves in horizontal and inclined rectangular channels with and without co-current air flow. In this work it was found that the "roll" wave transition was a function of both liquid and gas rates and, by adapting the theory of Jeffreys [16], the theoretical prediction of the transition point was possible. In discussing the work of Knuth, Hanratty and Hershman point out that in horizontal flow an asymetry in the effect of gravity occurs which makes the results difficult to interpret. They suggest, qualitatively, that one might expect at high gas Reynolds numbers that the liquid Reynolds number for transition would become more or less independent of the gas rate. This qualitative observation is certainly in accord with the present results (see Fig. 4). It would seem worth while to develop a theory on similar lines to that of Hanratty and Hershman applicable to upwards vertical flow; one difficulty would be, however, to distinguish clearly what one means by "roll" waves. At first sight the "disturbance" waves discussed in the present report might be expected to correspond to "roll" waves as defined by Hanratty and Hershman. These authors discuss the application of their theory to waves on falling films; some of the so-called "ripple" waves observed in the present paper are similar to those observed on falling films and there might be an unfortunate confusion of terms here. The "disturbance" waves, however, have a very characteristic appearance and behaviour and may arise from a quite different mechanism from the roll waves of HANRATTY and HERSHMAN.

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Résumé—L'écoulement de deux phases dans un espace annulaire va presque toujours de pair avec la formation d'ondes qui se propagent rapidement à la surface du film liquide et présentent un aspect laiteux, causé par la diffraction de la lumière sur leur surface irrégulière. On a étudié les conditions sous lesquelles ces ondes se forment et on a mesuré leur vitesse, séparation et fréquence. La formation des ondes semble coincider avec une transition dans l'entrainement observé dans un travail précédent. Les mesures du mouvement des films faites à l'aide d'une caméra sont en accord avec celles basées sur la conductibilité. La vitesse moyenne des ondes est de 4–10 ft/sec, tandis que la vitesse de l'air est de 40–150 ft/sec et la vitesse moyenne de l'eau dans le film de 1 à 3 ft/sec. La vitesse moyenne des ondes augmente rapidement avec le débit d'air mais elle est insensible aux variations du débit de l'eau. La fréquence des ondes, par contre, est insensible au débit d'air, mais augmente proportionnellement avec le débit du liquide ce qui suggère que les ondes transportent le liquide. Dans tous les cas les vitesse des ondes ne sont pas distribuées également par rapport à la moyenne et l'histogramme présente un certain nombre de pics. Ces pics apparaissent aux mêmes positions pour différents débits ce qui pourrait impliquer l'existence de "vitesses préférées". Des observations similaires, mais sujettes à équivoque, ont été faites en ce qui concerne la séparation des ondes.

Zusammenfassung—Beim Fliessen eines Flüssigkeitsfilms über Flächen von ringförmigem Querschnitt werden meistens grosse Störstellen beobachtet, die rasch wandern und infolge ihrer stark gewellten Oberfläche ein milchiges Aussehen aufweisen. Es wurden die Bedingungen untersucht, unter welchen diese Störstellen entstehen, und ihre Geschwindigkeit und Frequenz gemessen.

Die Messungen der Wellenbewegung erfolgte durch Filmen und Bestimmung der Leitfähigkeit. Beide Methoden ergaben übereinstimmende Resultate. Die mittlere Geschwindigkeit der Störwellen betrug 4–10 ft./sec., verglichen mit einer Luftgeschwindigkeit von 40–150 ft./sec. und einer mittleren Wassergeschwindigkeit im Flüssigkeitsfilm von 1–3 ft./sec. Die mittlere Wellengeschwindigkeit stieg mit wechselnder Luftgeschwindigkeit stark an, war aber unabhängig von der Wassergeschwindigkeit. Dagegen war die Wellenfrequenz unabhängig von der Luftgeschwindigkeit und stieg mit wechselnder Wassergeschwindigkeit an, was daraufhin deutet, dass die Wellen die Flüssigkeit transportieren. Die Wellengeschwindigkeit scheint nicht normal um den Mittelwert verteilt zu sein und es treten bevorzugte Wellengeschwindigkeiten auf.